7.5 Properties of Trapezoids and Kites

Name _____ Period ____ Date _____

Geometry 7.5 (30-minute plan) · Student sheet

I can…

- find the angle measures of an isosceles trapezoid.
- use the Trapezoid Midsegment Theorem to find lengths.
- find the angle measures of a kite and name a quadrilateral as specifically as possible.

Vocabulary · fill in the missing word

Trapezoid — a _____ with exactly one pair of parallel sides (the bases); the other sides are the legs

Isosceles trapezoid — a trapezoid with _____ legs

Base angles — a pair of angles that _____ a base

Midsegment of a trapezoid — the segment that joins the _____ of the legs

Kite — a _____ with two pairs of consecutive congruent sides, but opposite sides not congruent

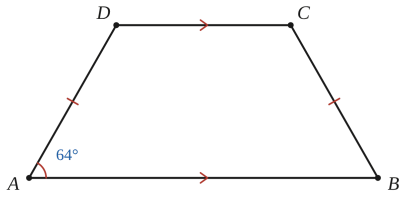
Part 1 · Notes and examples (fill in each note, then work the example)

Trapezoid: A **trapezoid** has exactly one pair of _____ sides (the bases).

Isosceles trapezoid: In an isosceles trapezoid, each pair of base angles is _____, and the diagonals are _____.

Example 1

ABCD is an isosceles trapezoid with bases $\overline{AB}$ and $\overline{CD}$, and $m\angle A = 64°$. Find $m\angle B$, $m\angle C$, and $m\angle D$.



Trapezoid Midsegment Theorem: the midsegment is _____ to the bases, and its length is _____ the sum of the bases.

Example 2

a. The bases of a trapezoid are 12 and 20. Find the length of the midsegment.

b. The midsegment of a trapezoid is 15 and one base is 11. Find the other base.

Kite: In a kite, the diagonals are _____, and exactly one pair of opposite angles is _____.

Example 3

In kite EFGH, the angles between the congruent sides measure 70° and 50°. Find the measure of each of the other two angles.

Part 2 · Partner practice

Partner practice

The midsegment of a trapezoid is 14. The bases are x + 4 and 3x. Find x and both bases.